**Practical No : 13**

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**Q.13) Write a C program for BFS.**

```
#include <stdio.h>
#define MAX 20
int graph[MAX][MAX];
int visited[MAX];
int queue[MAX];
int n;
void BFS(int start)
{
   int front = 0, rear = 0;
   int i, current;
   // Mark starting vertex as visited
   visited[start] = 1;
   queue[rear++] = start;
   while (front < rear) {
      current = queue[front++];
      printf("%d ", current + 1);
      // Visit all adjacent vertices
      for (i = 0; i < n; i++) {
         if (graph[current][i] == 1 && visited[i] == 0) {
            visited[i] = 1;
            queue[rear++] = i;
         }
      }
   }
}
```

```
int main()
{
    int i, j, start;
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter the adjacency matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);
        }
    }
    // Initialize all vertices as unvisited
    for (i = 0; i < n; i++) {
        visited[i] = 0;
    }
    printf("Enter starting vertex: ");
    scanf("%d", &start);
    printf("\nBFS Traversal: ");
    BFS(start - 1);
    printf("\n");
    return 0;
}
```

Output:

